

Department of Engineering Cybernetics
TTK4215 Adaptive Control
Solution 10

Problem 5 of Chapter 7 in Ioannou & Fidan We have

$$\Lambda(z) = z^{n-1} \quad (1)$$

$$W_m(z) = z^{-n^*} \quad (2)$$

which imply that $k_m = 1$, $Z_m = 1$, $R_m = z^{n^*}$ and $\Lambda_0(z) = z^{n-1}$. From the definition of $Q(z)$ following (7.26), we have

$$Q(z) = \sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1+n^*-n_p-i}. \quad (3)$$

From (7.24) and (7.26) we have

$$u_p = z^{-n+1} \left(z^{n-1} - \left(z^{m_p} + \sum_{i=0}^{m_p-1} b_i z^i \right) Q(z) \right) u_p \quad (4)$$

$$+ \frac{1}{k_p} z^{-n+1} \left(Q(z) \left(z^{n_p} + \sum_{i=0}^{n_p-1} a_i z^i \right) - z^{n-1+n^*} \right) y_p \quad (5)$$

$$+ \frac{1}{k_p} z^{n^*} y_m \quad (6)$$

$$= z^{-n+1} \left(z^{n-1} - \left(z^{m_p} + \sum_{i=0}^{m_p-1} b_i z^i \right) Q(z) \right) u_p \quad (7)$$

$$+ \frac{1}{k_p} z^{-n+1} \left(Q(z) \left(z^{n_p} + \sum_{i=0}^{n_p-1} a_i z^i \right) - z^{n-1+n^*} \right) y_p \quad (8)$$

$$+ \frac{1}{k_p} z^{n^*} y_m \quad (9)$$

Consider the terms in u_p on the right hand side. We have

$$\begin{aligned} & z^{-n+1} \left(z^{n-1} - \left(z^{m_p} + \sum_{i=0}^{m_p-1} b_i z^i \right) Q(z) \right) u_p \\ &= z^{-n+1} \left(z^{n-1} - \left(z^{m_p} \sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1+n^*-n_p-i} + Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) \right) u_p \\ &= z^{-n+1} \left(z^{n-1} - \left(\sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1-i} + Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) \right) u_p \\ &= -z^{-n+1} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1-i} + Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) u_p \\ &= - \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{-i} + z^{-n+1} Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) u_p \quad (10) \end{aligned}$$

Notice that the b 's used above as defined for the MRC in section 7.2.2, are related to the b 's defined in section 7.3 (see eq. (7.54)) through the constant

k_p . The expression above is therefore equivalent to

$$- \frac{1}{b_m} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{-i} + z^{-n+1} Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) u_p \quad (11)$$

where the b 's are now as defined in (7.54). Similarly for the terms in y_p on the right hand side (using $k_p = b_m$)

$$\begin{aligned} & \frac{1}{b_m} z^{-n+1} \left(Q(z) \left(z^{n_p} + \sum_{i=0}^{n_p-1} a_i z^i \right) - z^{n-1+n^*} \right) y_p \\ &= \frac{1}{b_m} z^{-n+1} \left(\left(z^{n_p} \sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1+n^*-n_p-i} + Q(z) \sum_{i=0}^{n_p-1} a_i z^i \right) - z^{n-1+n^*} \right) y_p \\ &= \frac{1}{b_m} z^{-n+1} \left(\left(\sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1+n^*-i} + Q(z) \sum_{i=0}^{n_p-1} a_i z^i \right) - z^{n-1+n^*} \right) y_p \\ &= \frac{1}{b_m} z^{-n+1} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n-1+n^*-i} + Q(z) \sum_{i=0}^{n_p-1} a_i z^i \right) y_p \\ &= \frac{1}{b_m} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n^*-i} + z^{-n+1} Q(z) \sum_{i=0}^{n_p-1} a_i z^i \right) y_p \quad (12) \end{aligned}$$

We have that

$$z^{-n+1} Q(z) = \sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n^*-n_p-i} \quad (13)$$

so (11) becomes

$$\begin{aligned} & - \frac{1}{b_m} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{-i} + z^{-n+1} Q(z) \sum_{i=0}^{m_p-1} b_i z^i \right) u_p \\ &= - \frac{1}{b_m} \left(\sum_{i=1}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{-i} + \sum_{i=0}^{n-1+n^*-n_p} (-a_{n_p-1})^i z^{n^*-n_p-i} \sum_{i=0}^{m_p-1} b_i z^i \right) u_p \\ &= - \frac{1}{b_m} \left(\sum_{i=1}^{n-1-m_p} (-a_{n_p-1})^i z^{-i} + \sum_{j=0}^{m_p-1} \sum_{i=0}^{n-1-m_p} (-a_{n_p-1})^i b_j z^{-m_p-i+j} \right) u_p \quad (14) \end{aligned}$$

Problem 9 of Chapter 7 in Ioannou & Fidan

a) All assumptions as specified in the lecture notes or book

b)

$$y_p = k_p \frac{Z_p}{R_p} u_p, \quad (15)$$

$$y_m = k_m \frac{Z_m}{R_m} r, \quad (16)$$

where $k_p = b_1$, $Z_p = z + b_0/b_1$, $R_p = 1^2 + a_1 z + a_0$, $k_m = 1$, $Z_m = 1$, and $R_m = z + 1$. Consider the control law

$$u_p = \theta_1^T[k] \omega_1[k] u_p + \theta_2^T[k] \omega_2[k] + \theta_3[k] y_p + c_0[k] r[k], \quad (17)$$

where $\Lambda = z + 1$. This control law can be implemented as

$$\omega_1[k+1] = -F \omega_1[k] + g u_p[k], \quad (18)$$

$$\omega_2[k+1] = -F \omega_2[k] + g y_p[k], \quad (19)$$

$$u_p = \theta^T[k] \omega[k], \quad (20)$$

where F and g are given by:

$$F = \begin{bmatrix} -\lambda_{n-2} & -\lambda_{n-3} & -\lambda_{n-4} & \cdots & -\lambda_0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \quad g = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

λ_i are the coefficients of

$$\Lambda(s) = s^{n-1} + \lambda_{n-2} s^{n-2} + \cdots + \lambda_1 s + \lambda_0 = \det(sI - F),$$

Hence a choice of $\Lambda = s + 1$ and $g = 1$ gives:

$$\omega_1[k+1] = -\omega_1[k] + u_p[k], \quad (21)$$

$$\omega_2[k+1] = -\omega_2[k] + y_p[k], \quad (22)$$

$$u_p = \theta^T[k] \omega[k], \quad (23)$$

where

$$\omega[k] = [\omega_1[k], \omega_2[k], y_p[k], u_p[k]]^T, \theta[k] = [\theta_1[k], \theta_2[k], \theta_3[k], c_0[k]]^T \quad (24)$$

c) An example for direct MRAC scheme would be to let $e = \frac{z[k] - \theta_P^T \phi[k]}{m_s^2[k]}$. The control law is then

$$u_p[k] = \theta^T[k] \omega[k] m_s^2[k] = 1 + \phi^T[k] * \phi[k], \quad (25)$$

$$\bar{\theta}_p[k+1] = \bar{\theta}_p[k] + \gamma \epsilon[k] \bar{\phi}[k], \quad (26)$$

$$\bar{c}_0[k+1] = c_0[k] + \gamma \epsilon[k] y_p[k], \quad (27)$$

$$\hat{c}_0[k+1] = \bar{c}_0[k+1] i f \bar{c}_0[k+1] sgn(c_0^*) \geq \beta_0, \quad (28)$$

$$\hat{k}_p = \beta_0 sgn(c_0^*) otherwise, \quad (29)$$

$$(30)$$

where

$$\bar{\omega} = [\omega_1^T[k], \omega_2^T[k], y_p^T[k]], \quad \bar{\phi}_p = W_m[z] \bar{\omega}, \quad (31)$$

$$\bar{\theta}[k] = [\theta_1^T[k], \theta_2^T[k], \theta_3^T[k]] \quad (32)$$